

Q1

26 ns

Setup $T_{clk} + T_{skew} - T_{jitter} \geq T_{cq,max} + T_{comb,max} + T_{su}$

Required Time = 25 + 3 - 2 = 26 ns

25 ns

in the path ~~above~~ Above

Arrival Time = $T_{cq,max} + T_{comb,max} = 6 + 15 = 21 ns$

• No setup Violation

27 ns

in the path below $26 ns \{ T_{clk} + T_{skew} - T_{jitter} \geq T_{cq,max} + T_{comb,max} + T_{su}$

Arrival time = $T_{cq,max} + T_{comb,max} = 6 + 17 = 23 ns$

There's a setup Violation ~~below~~ : Slack = -1 ns

5 ns

Hold $T_{cq,min} + T_{comb,min} \geq T_{hold} + T_{skew} + T_{jitter}$

Arrival time = $T_{cq,min} + T_{comb,min} = 2 + 3 = 5 ns$

8 ns

(above)

There's a Hold Violation : Slack = -3 ns

below Arrival time = 9 ns : No Hold Violation

view

Q2

1) * Bottom path $\Rightarrow$ Reduce Comp Logic ~~Delay~~ Delay

* Reduce f $\odot$ Increase Tclk

* pipelining

2) Top path $\Rightarrow$ * Insert Delay Buffers * Clock tree Balance